Experiment 24: Dialog Act Recognition in Python

1. Aim

To write a Python program that recognizes dialog acts in a given dialog or conversation.

2. Procedure

1. Define a function to classify conversational utterances into dialog act categories.
2. Check for question marks, greeting words, and farewell phrases in each utterance.
3. Define a sample conversation containing multiple turns.
4. Apply the dialog act recognition function to each utterance.
5. Display each utterance along with its recognized dialog act.

3. Program

```
def get_dialog_act(text):
    if text.endswith("?"): return "Question"
    if text.lower().startswith(("hi", "hello")): return "Greeting"
    if text.lower().startswith(("bye", "thanks")): return "Closing"
    return "Statement"

dialog = ["Hello!", "How are you?", "I am learning NLP.", "Thanks, bye!"]
for turn in dialog:
    print(f"{turn} -> {get_dialog_act(turn)}")
```

4. Output

```
Hello! -> Greeting
How are you? -> Question
I am learning NLP. -> Statement
Thanks, bye! -> Closing
```

5. Result

Thus, the Python program was successfully executed to recognize dialog acts in a given conversation.